Musanze Harvest Intelligence Service

Technical report: an individual reproducible machine learning system and API

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Verification code	**AI-A2-7389**
Personal seed (LLLL)	7389
Programme / module	Software Engineering, Year 3 / SWE 3513 Artificial Intelligence
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1. Identity and decision problem

1.1 Stakeholder

The stakeholder is the field officer of a potato cooperative in Musanze. The cooperative receives farm records in inconsistent spreadsheets and has no evidence-based way to plan visits. Internet access may be unreliable, so training and command-line prediction run fully offline, and a small API is provided for a future mobile application.

1.2 Prediction questions

Question for the field officer	ML task	Target	Output of the system
How many tons will this farm harvest?	Regression	yield_tons	Yield estimate (tons)
Is this farm at high production risk?	Classification	high_risk (yield per hectare below the median)	Prediction, probability and risk band
Which farms have similar conditions?	Clustering (K-Means)	none (inputs only)	Cluster number and plain-language profile

1.3 Ethical boundary

This is a classroom decision-support prototype trained on **synthetic** data. It does not replace an agronomist and must never be used to decide a farmer's access to finance, insurance or public services. Every prediction carries *human_review_required = true*, and the API returns the synthetic-data notice with each response. Inputs outside the training ranges are rejected instead of extrapolated.

2. Data and reproducibility

2.1 Data dictionary

Column	Type	Generation rule / meaning
farm_id	text	Row identifier F001... (not a model input)
rainfall_mm	numeric	Uniform 600 to 1400
avg_temp_c	numeric	Uniform 12 to 26
soil_moisture_pct	numeric	Uniform 20 to 80 (12 values deliberately missing)
fertilizer_kg	numeric	Uniform 0 to 250 (12 values deliberately missing)
field_size_ha	numeric	Uniform 0.2 to 4.0
season	categorical	Random A or B
cooperative_support	binary	Random 0 or 1
yield_tons	target (regression)	0.004 rainfall + 0.06 soil moisture + 0.02 fertilizer + 1.30 field size - 0.25 |temp - 18| + 0.70 support + season effect (A +0.50, B -0.20) + Normal(0, 0.80) noise; clipped at 0.50
high_risk	target (classification)	1 if yield per hectare is below its median, else 0

2.2 Personal seed rule and size

The registration number 25/27389 gives LLLL = **7389**. The generator uses `numpy.random.default_rng(7389)`. Base rows = 240 + (7389 mod 41) = 240 + 9 = **249**. After appending three exact duplicates the file has **252** rows (`data/AI_A2_2527389.csv`, numbers saved with four decimals). run_all.py rejects any seed other than 7389.

2.3 Deliberate data problems and how they were handled

Problem	Evidence in this run	Handling
Missing soil_moisture_pct	12 rows (4.76%)	Median imputation fitted on training rows only (inside the Pipeline)
Missing fertilizer_kg	12 rows (4.76%)	Same as above
Exact duplicate rows	3 rows	Removed before splitting: 252 to 249 rows, so copies cannot leak across train and test
Schema, ranges, labels	All checks passed	Validation rejects missing columns, out-of-range values and invalid labels

2.4 Dataset fingerprint (SHA-256 of the final CSV bytes)

```
8acc1963f35734be0573f8f05034bf944c451a4a02297a837060250cc81d61e5
```

The same fingerprint was produced on two different machines (Linux with Python 3.12, and the author's Windows computer with Python 3.13) and is written to `artifacts/data_fingerprint.txt` on every run. Re-running run_all.py reproduces the fingerprint and, within floating-point tolerance, the metrics (this is covered by an automated test).

3. Pipeline design

3.1 Flow from raw file to saved models

Step	What happens	Module
1 Generate	Create the CSV from the personal seed and write the SHA-256 fingerprint	data/generate_data.py
2 Validate	Check columns, ranges, labels, missing values, duplicates and leakage	src/validation.py
3 Clean	Drop exact duplicates before any splitting	src/preprocessing.py
4 Split	80/20 split, random_state = 7389, stratified on high_risk: 199 training rows, 50 test rows	src/preprocessing.py
5 Preprocess	Feature builder, then median imputer + standard scaler (numeric), one-hot encoder (season), most-frequent imputer (binary), all in one Pipeline	src/preprocessing.py
6 Model	Baselines and candidates for regression and classification, selection by 5-fold cross-validation on the training split	src/modeling.py
7 Cluster	Scale inputs, test K = 2 to 6, choose K by silhouette score, write profiles	src/clustering.py
8 Save and serve	Save pipelines with joblib; the CLI (predict.py) and the API (app.py) load the same files	src/inference.py

3.2 Leakage controls

- **Targets are never inputs.** yield_tons, yield per hectare, high_risk and farm_id are blocked by a leakage guard that stops training if any of them appears among the features.
- **Train-only fitting.** Imputers, scaler and encoder are steps of a scikit-learn Pipeline, so they are fitted on training rows only, inside each cross-validation fold, and applied unchanged to the test set and to new records.
- **Duplicates removed before the split**, so an identical farm cannot appear on both sides.
- **The test set is used only for the final report.** Models were chosen with training-only cross-validation.
- **Clustering never sees the targets.** Yield and risk appear in the cluster profiles only after the clusters are formed, and are labelled as after-the-fact.

3.3 Engineered features and model candidates

Three features are computed from each row's own inputs: *temp_gap_c* = |avg_temp_c - 18| (the formula's optimum temperature), *fertilizer_per_ha* = fertilizer_kg / field_size_ha and *water_index* = rainfall_mm x soil_moisture_pct / 1000.

Task	Baseline	Candidates
Regression	Mean predictor	Ridge regression, random forest (150 trees), gradient boosting
Classification	Majority class	Logistic regression, random forest (150 trees), gradient boosting
Clustering	-	K-Means with K = 2, 3, 4, 5, 6 (n_init = 10)

Every random state is set to 7389 (split, cross-validation folds, forests, boosting, K-Means, PCA), and models use a single thread so results are deterministic.

4. Regression and classification

4.1 Regression: held-out test set (50 farms)

Model	CV RMSE (train)	MAE (t)	RMSE (t)	R^2
Mean baseline	-	2.254	2.895	0.000
Ridge regression (selected)	**0.991**	**0.789**	**0.992**	**0.883**
Random forest	1.524	1.307	1.575	0.704
Gradient boosting	1.387	1.202	1.438	0.753

Selected: Ridge regression (lowest cross-validated RMSE, 0.991 t). The data formula is linear, so a linear model fits it best; the tree models are noisier with only about 200 training rows. The simulated noise has standard deviation 0.80 t, which sets a floor near 0.80 t for any model's RMSE, so Ridge (RMSE 0.992 t) is already close to the best achievable.

4.2 Classification: held-out test set (25 not high, 25 high risk)

Model	CV F1 (train)	Precision	Recall	F1	ROC-AUC
Majority baseline	-	0.000	0.000	0.000	0.500
Logistic regression (selected)	**0.969**	**0.893**	**1.000**	**0.943**	**0.986**
Random forest	0.941	0.889	0.960	0.923	0.981
Gradient boosting	0.917	0.889	0.960	0.923	0.965

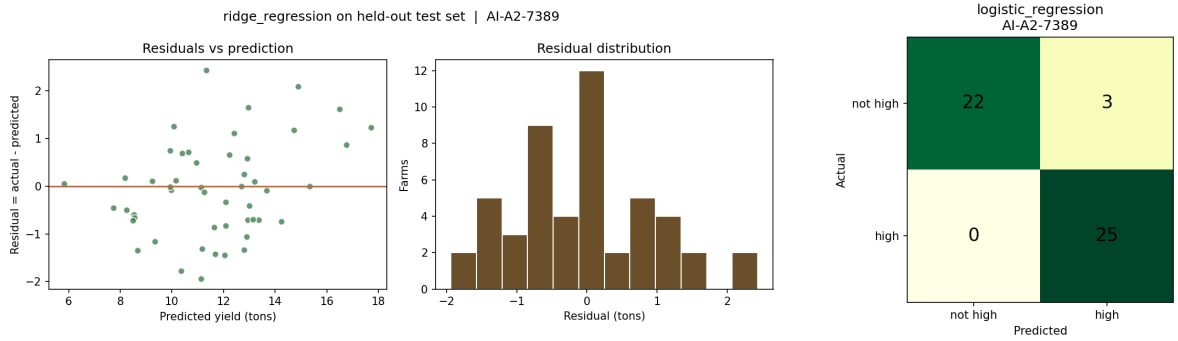


Figure 1: residuals of the selected regression model (left) and confusion matrix of the selected classifier (right).

Reading the plots. Residuals scatter evenly around zero with no curve or funnel, so Ridge is unbiased across the prediction range. The confusion matrix has 22 true negatives, 3 false positives, 0 false negatives and 25 true positives: recall is 1.000 (no high-risk farm was missed in the test set) and precision is 0.893 because of 3 false alarms. The majority baseline scores F1 = 0 because it never predicts high risk, which shows why the baseline matters.

4.3 One important regression error (farm F205)

Farm F205 was under-predicted by **2.44 t** (actual 13.77 t, predicted 11.33 t), the largest error in the test set. Its *fertilizer_kg* value was missing, so the pipeline substituted the training median (about 121 kg). In the data formula each extra 100 kg of fertilizer adds 2 t, so if this farm really used a lot of fertilizer the imputed value would explain most of the gap; random noise (standard deviation 0.80 t) may explain the rest. The model cannot tell which from the record alone. Lesson: predictions for farms with missing fertilizer or soil-moisture values deserve extra caution and should be re-checked by the field officer.

4.4 One important classification error (farm F188)

Farm F188 was predicted high risk with probability **0.81** but is not high risk. It has a large field (3.43 ha), and larger fields lower yield per hectare in this data. Its true yield per hectare is 5.53 t/ha while the median threshold is 5.48 t/ha, so it sits only just above the line. The other two most confident mistakes (F200 at 5.60 and F185 at 5.49 t/ha) are also within 0.13 t/ha of the threshold. The errors are therefore boundary cases and not a systematic failure, and the cost of a false alarm is only an extra field visit. With 50 test farms, these figures are indicative, not precise.

5. Clustering

5.1 Choosing K

K	2	3	4	5	6
Silhouette score	0.1405	0.1320	0.1386	0.1459	0.1408

Inputs were median-imputed and standardised, with season one-hot encoded and no target used. **K = 5** has the highest silhouette score (0.1459). All five scores lie between 0.13 and 0.15, so the differences are small and the groups are weak tendencies, not sharply separated clusters. This is expected, because every feature was drawn independently and there is no natural group structure in the data. K = 5 is useful as a practical grouping but should not be over-interpreted.

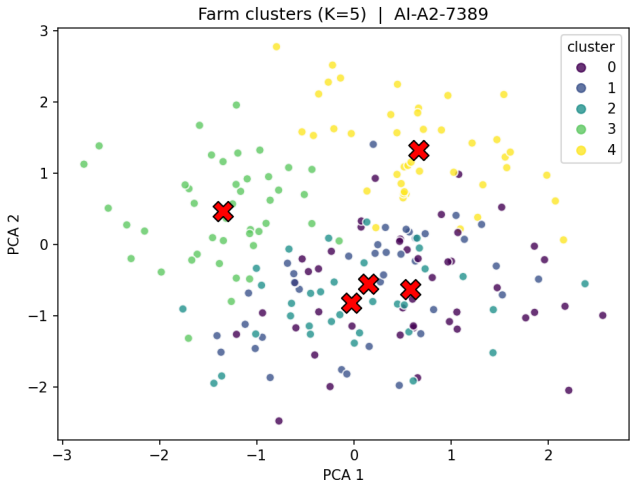


Figure 2: the five clusters on the first two principal components (training farms; crosses are cluster centres).

5.2 Cluster profiles (training farms)

Cl.	Farms	Plain-language profile	Rain (mm)	Temp (C)	Soil (%)	Fert (kg)	Field (ha)	High-risk share*
0	44	wetter, warmer, lower soil moisture	1144	22.5	38.7	112	1.75	36%
1	37	drier, warmer, heavy fertilizer use	820	22.8	53.2	181	1.82	40%
2	33	wetter, cooler, higher soil moisture, smaller fields	1173	16.0	61.7	125	1.15	3%
3	43	drier, higher soil moisture, light fertilizer use, larger fields	810	18.2	60.9	69	2.69	77%
4	42	cooler, lower soil moisture, larger fields	993	15.4	31.3	144	2.92	81%

* Share of farms in the cluster that are high risk. This uses the target after the clusters were formed and was not used to form them.

5.3 One farm cluster explained: Cluster 3

Cluster 3 holds 43 training farms. They are drier than average (about 810 mm of rain), have higher soil moisture, use light fertilizer (about 69 kg) and have the larger fields (2.69 ha on average). After the fact, 77% of them are high risk, much higher than the 50% overall. The reason is that high risk is measured per hectare: a large field with little rain and little fertilizer spreads its yield over many hectares. **Operational meaning:** these are the farms to visit first. Fertilizer advice and water management look like the most useful help here. Cluster 4 (cooler, low soil moisture, larger fields, 81% high risk) deserves the same priority, while Cluster 2 (wetter, cooler, smaller fields, 3% high risk) needs only routine monitoring.

6. API and testing

6.1 Endpoint contract

Endpoint	Purpose	Success	Failure
GET /health	Service and model status, verification code	200	-
GET /model-info	Seed, fingerprint, selected models, test metrics, input ranges, notice	200	503 if models are not trained
POST /predict	Typed farm record in, yield, risk, probability, cluster and verification code out	200	422 invalid input (field-by-field list); 503 models missing

All seven fields are required, numbers are range-checked, season must be A or B, support must be 0 or 1, and unknown fields (such as yield_tons) are rejected.

6.2 Valid request and response

```
POST /predict with {"rainfall_mm": 1020.5, "avg_temp_c": 17.8, "soil_moisture_pct": 55.0,
"fertilizer_kg": 120.0, "field_size_ha": 1.6, "season": "A", "cooperative_support": 1}

Response 200: {"verification_code": "AI-A2-7389", "yield_estimate_tons": 13.146,
"high_risk_prediction": false, "high_risk_probability": 0.0272, "risk_band": "low", "cluster": 2,
"cluster_profile": "Cluster 2: wetter, cooler, higher soil moisture, smaller fields.",
"human_review_required": true}
```

6.3 Invalid request and response

```
POST /predict with rainfall_mm = 2500 (all other fields valid)

Response 422: {"error": "invalid_input", "problems": [{"field": "rainfall_mm", "problem": "Input should
be less than or equal to 1400", "type": "less_than_equal"}]}
```

The command-line tool applies the same validation: python predict.py --json samples/invalid.json prints the problems and exits with code 2.

6.4 Test summary

19 tests, 19 passed (`python -m pytest -q`, 1 harmless library deprecation warning). The suite covers failure paths as well as the happy path:

Area	What is tested
Inference	Valid record returns every field; missing field, wrong type, five range/category errors and an unknown field are all rejected
Identity and data	Wrong seed rejected; row count follows 240 + (LLLL mod 41); missing values and duplicates detected and cleaned; leakage guard blocks target columns
Reproducibility	Running the whole pipeline twice gives the same fingerprint and the same metrics
API and CLI	/health, /model-info, valid /predict, invalid /predict gives 422 naming the fields, no trained models gives 503, CLI exits with code 2 on invalid input

The full test list is in the separate test log (AI_A2_2527389_TEST_LOG.pdf).

7. Model card and reflection

Item	Statement
Intended use	Classroom decision-support prototype that helps a field officer estimate yield, spot farms at high production risk and group similar farms for planning visits.
Data status	Entirely synthetic, generated from a known formula and a personal seed. It contains no real farm, person or location data.
Limitations	About 199 training rows; features are independent and the relationships are exactly the formula, so real farms will be messier. Test metrics come from only 50 farms. Clusters are weak (silhouette about 0.15). Inputs are limited to the generator's ranges, and imputed values reduce accuracy.
Risks	Over-trusting a single number; misreading weak clusters as real farm types; treating a false alarm or a missed risk as a judgement about a farmer.
Prohibited use	Deciding access to finance, insurance or public services; replacing an agronomist; use on real farms without retraining and independent validation.
Human oversight	Every output carries human_review_required = true and a notice. A person must review each result and may override it.

7.1 What I would improve with real data

- Collect real farm records with a consistent form to remove the spreadsheet inconsistencies, and record the real reason for any missing value.
- Retrain and compare non-linear models properly, since real relationships are rarely linear, and check performance for each season and support group to look for unfair differences.
- Add uncertainty (prediction intervals) so the officer sees how sure the model is, and monitor the model for drift each season.
- Validate the clusters with agronomists, and replace the median-based risk label with a threshold that the cooperative itself defines.

7.2 Reflection and AI-use note

The main lesson was how much of a good result comes from the pipeline rather than the algorithm: removing duplicates before the split, fitting preprocessing on training data only, and choosing models by cross-validation. The surprise was that the simplest model (Ridge) won, which is a reminder to match the model to the data. The system was built with the help of an AI assistant (Claude, by Anthropic); this is disclosed in evidence/AI_USE.md and in the signed declaration. I ran the system on my own computer and the results in this report were generated by it.